

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

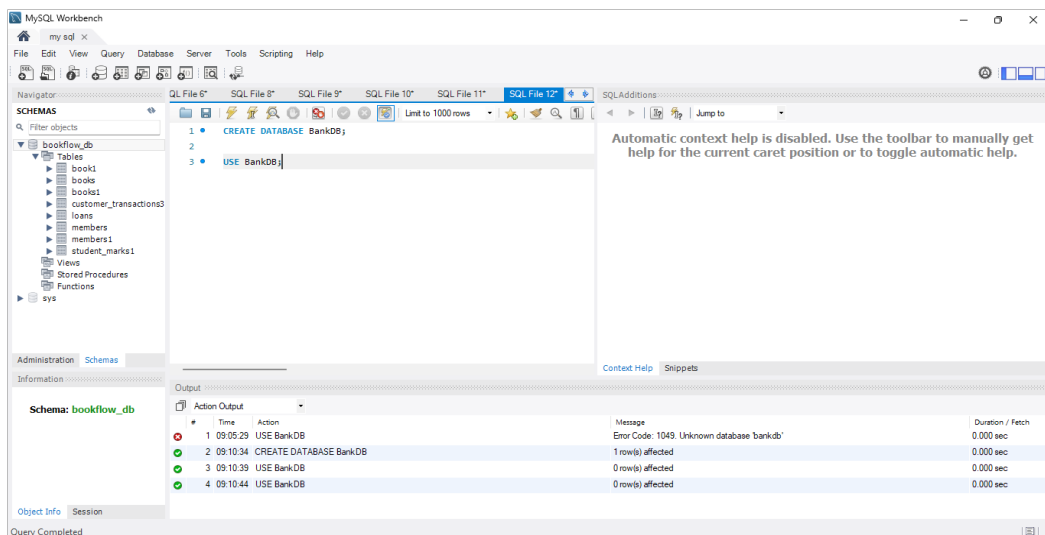
Sec:- 3

Name:- Srithika Kambhammettu

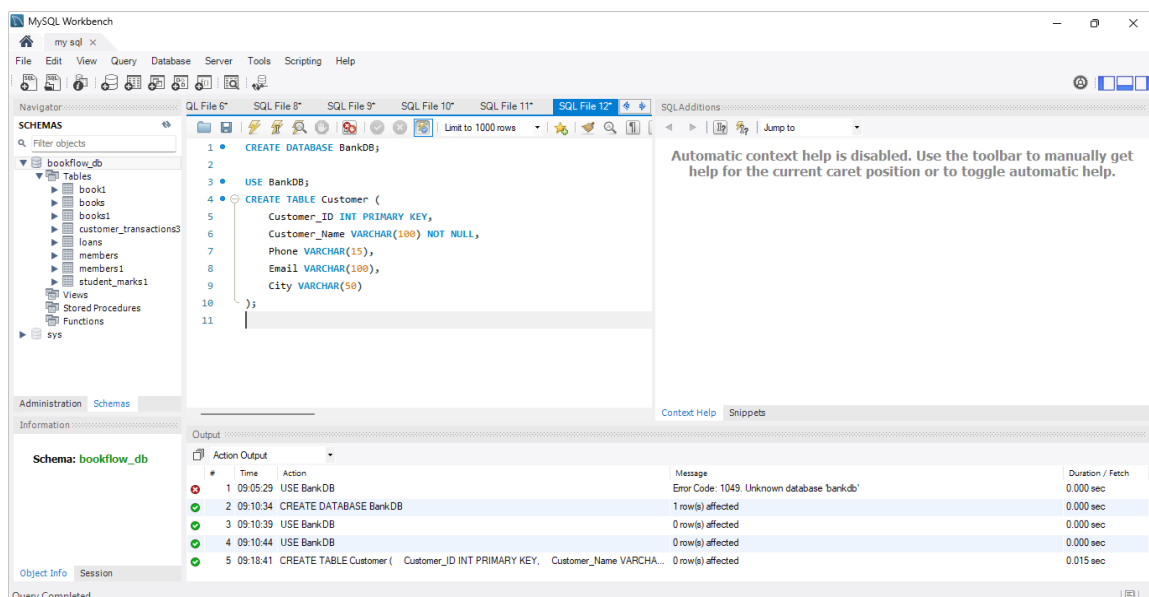
Roll no:- 2520030361

6 - Topic: Bank Database – Triggers and Stored Procedures

1. CREATE DATABASE



1. 2. CREATE CUSTOMER TABLE



3. CREATE ACCOUNT TABLE

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected in the Navigator. The SQL editor contains the following code:

```
7 Phone VARCHAR(15),
8 Email VARCHAR(50) Stop the query being executed (the connection to the DB server will not be restarted and any open transactions will remain open) on or to toggle automatic help.
9 City VARCHAR(50)
10 );
11 CREATE TABLE Account (
12 Account_No INT PRIMARY KEY,
13 Customer_ID INT,
14 Account_Type VARCHAR(20),
15 Balance DECIMAL(12,2) DEFAULT 0,
16 Branch VARCHAR(50),
17 FOREIGN KEY (Customer_ID)
18 REFERENCES Customer(Customer_ID)
19 );
20
21
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
2	09:10:34	CREATE DATABASE BankDB	1 row(s) affected	0.000 sec
3	09:10:39	USE BankDB	0 row(s) affected	0.000 sec
4	09:10:44	USE BankDB	0 row(s) affected	0.000 sec
5	09:18:41	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(50),	0 row(s) affected	0.015 sec
6	09:26:51	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Acc...	0 row(s) affected	0.047 sec

4. CREATE TRANSACTION TABLE

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected in the Navigator. The SQL editor contains the following code:

```
18 FOREIGN KEY (Customer_ID)
19 REFERENCES Customer(Customer_ID)
20 );
21
22 CREATE TABLE Bank_Transaction (
23 Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
24 Account_No INT,
25 Transaction_Type VARCHAR(20),
26 Amount DECIMAL(12,2),
27 Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
28
29 FOREIGN KEY (Account_No)
30 REFERENCES Account(Account_No)
31 );
32
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
3	09:10:39	USE BankDB	0 row(s) affected	0.000 sec
4	09:10:44	USE BankDB	0 row(s) affected	0.000 sec
5	09:18:41	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(50),	0 row(s) affected	0.015 sec
6	09:26:51	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Acc...	0 row(s) affected	0.047 sec
7	09:35:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREM...	0 row(s) affected	0.016 sec

5. CREATE LOAN TABLE

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected in the Navigator. The SQL Editor contains the following SQL code:

```
28
29
30 FOREIGN KEY (Account_No)
31 REFERENCES Account(Account_No)
32 );
33
34 CREATE TABLE Loan (
35 Loan_ID INT PRIMARY KEY,
36 Customer_ID INT,
37 Loan_Type VARCHAR(30),
38 Loan_Amount DECIMAL(12,2),
39 Interest_Rate DECIMAL(5,2),
40
41 FOREIGN KEY (Customer_ID)
42 REFERENCES Customer(Customer_ID)
43 );
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
4	09:10:44	USE BankDB	0 row(s) affected	0.000 sec
5	09:18:41	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(50), Customer_Phone VARCHAR(15), Customer_Email VARCHAR(50), Customer_City VARCHAR(50))	0 row(s) affected	0.015 sec
6	09:26:51	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Amount DECIMAL(12,2), Account_Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.047 sec
7	09:35:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Customer_ID INT, Transaction_Amount DECIMAL(12,2), Transaction_Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.016 sec
8	09:36:44	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount DECIMAL(12,2), Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.015 sec

6. INSERT CUSTOMER DATA

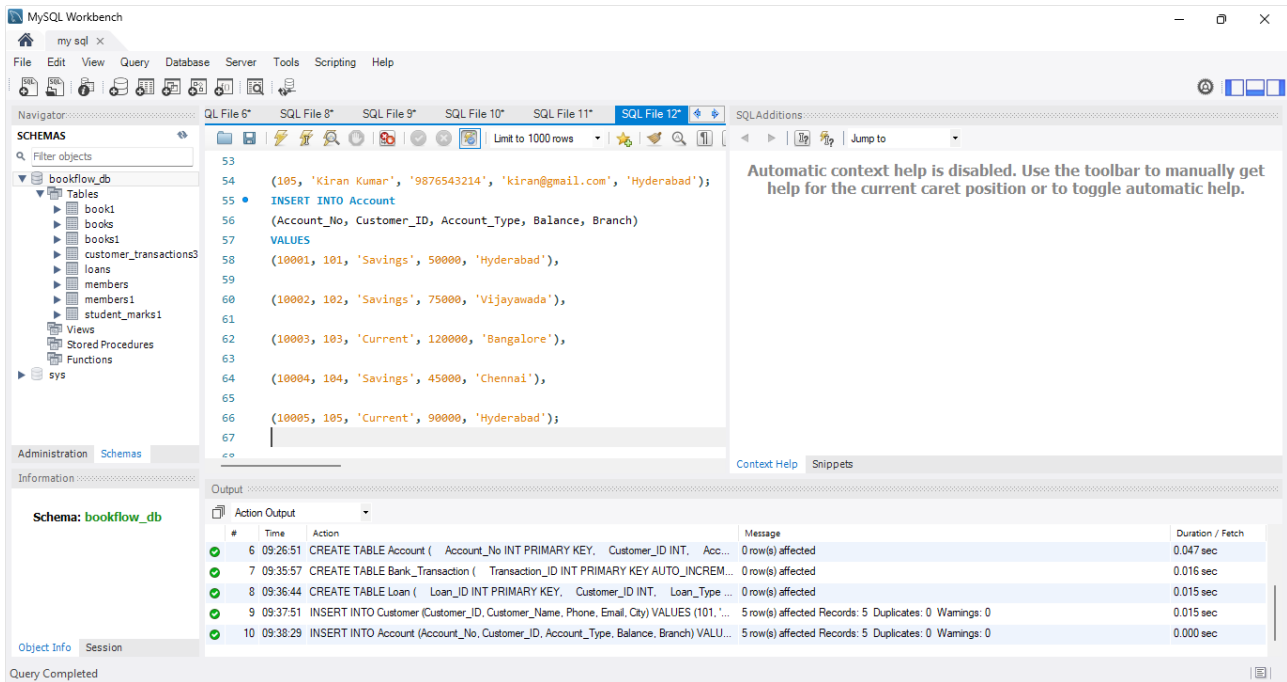
The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL Editor contains the following SQL code:

```
41
42
43 INSERT INTO Customer
44 (Customer_ID, Customer_Name, Phone, Email, City)
45 VALUES
46 (101, 'Srithika Kambhammettu', '9876543210', 'srithika@gmail.com', 'Hyderabad'),
47 (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),
48 (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
49 (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
50 (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
5	09:18:41	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(50), Customer_Phone VARCHAR(15), Customer_Email VARCHAR(50), Customer_City VARCHAR(50))	0 row(s) affected	0.015 sec
6	09:26:51	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Amount DECIMAL(12,2), Account_Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.047 sec
7	09:35:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Customer_ID INT, Transaction_Amount DECIMAL(12,2), Transaction_Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.016 sec
8	09:36:44	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount DECIMAL(12,2), Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.015 sec
9	09:37:51	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Srithika Kambhammettu', '9876543210', 'srithika@gmail.com', 'Hyderabad'), (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'), (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'), (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'), (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec

7. INSERT ACCOUNT DATA



MySQL Workbench interface showing SQL File 12* with the following queries:

```
53
54 (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
55 INSERT INTO Account
56 (Account_No, Customer_ID, Account_Type, Balance, Branch)
57 VALUES
58 (10001, 101, 'Savings', 50000, 'Hyderabad'),
59
60 (10002, 102, 'Savings', 75000, 'Vijayawada'),
61
62 (10003, 103, 'Current', 120000, 'Bangalore'),
63
64 (10004, 104, 'Savings', 45000, 'Chennai'),
65
66 (10005, 105, 'Current', 90000, 'Hyderabad');
67
```

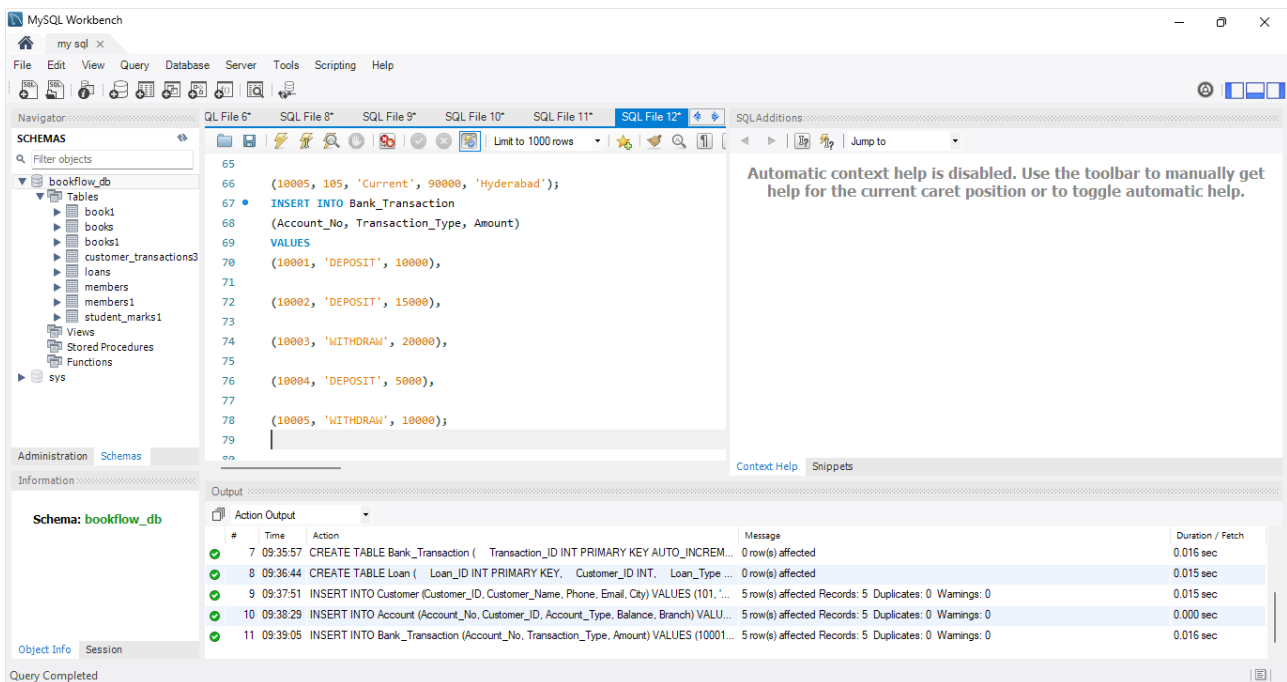
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output:

#	Time	Action	Message	Duration / Fetch
6	09:26:51	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Acc...	0 row(s) affected	0.047 sec
7	09:35:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREM...	0 row(s) affected	0.016 sec
8	09:36:44	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type ...	0 row(s) affected	0.015 sec
9	09:37:51	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	09:38:29	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALU...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec

Query Completed

8. INSERT TRANSACTION DATA



MySQL Workbench interface showing SQL File 12* with the following queries:

```
65
66 (10005, 105, 'Current', 90000, 'Hyderabad');
67 INSERT INTO Bank_Transaction
68 (Account_No, Transaction_Type, Amount)
69 VALUES
70 (10001, 'DEPOSIT', 10000),
71
72 (10002, 'DEPOSIT', 15000),
73
74 (10003, 'WITHDRAW', 20000),
75
76 (10004, 'DEPOSIT', 5000),
77
78 (10005, 'WITHDRAW', 10000);
79
80
```

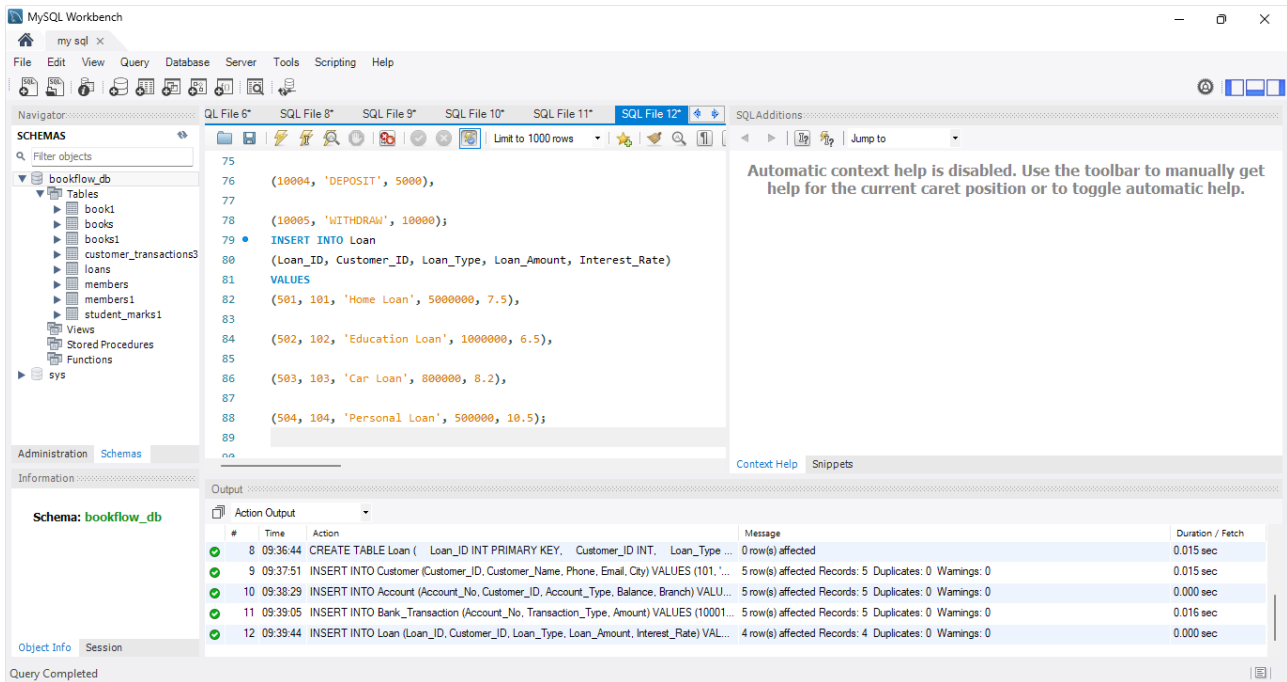
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output:

#	Time	Action	Message	Duration / Fetch
7	09:35:57	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREM...	0 row(s) affected	0.016 sec
8	09:36:44	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type ...	0 row(s) affected	0.015 sec
9	09:37:51	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	09:38:29	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALU...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	09:39:05	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

Query Completed

9. INSERT LOAN DATA



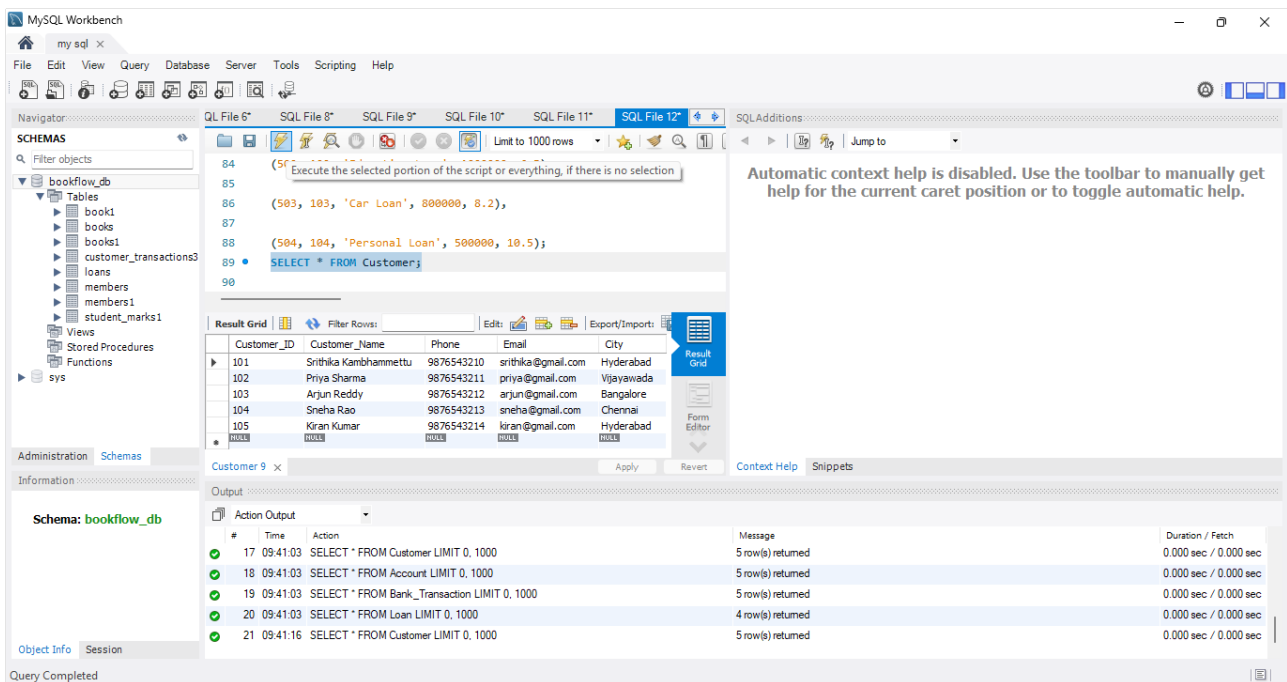
MySQL Workbench interface showing SQL File 12 with the following SQL statements:

```
75
76 (10004, 'DEPOSIT', 5000),
77
78 (10005, 'WITHDRAW', 10000);
79 • INSERT INTO Loan
80 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
81 VALUES
82 (501, 101, 'Home Loan', 5000000, 7.5),
83
84 (502, 102, 'Education Loan', 1000000, 6.5),
85
86 (503, 103, 'Car Loan', 800000, 8.2),
87
88 (504, 104, 'Personal Loan', 500000, 10.5);
89
na
```

Output pane showing the results of the queries:

#	Time	Action	Message	Duration / Fetch
8	09:36:44	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type ...	0 row(s) affected	0.015 sec
9	09:37:51	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	09:38:29	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALU...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	09:39:05	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
12	09:39:44	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VAL...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec

10. DISPLAY TABLE DATA



MySQL Workbench interface showing SQL File 12 with the following SQL statements:

```
84
85
86 (503, 103, 'Car Loan', 800000, 8.2),
87
88 (504, 104, 'Personal Loan', 500000, 10.5);
89 • SELECT * FROM Customer;
90
```

Output pane showing the results of the query:

Customer_ID	Customer_Name	Phone	Email	City
101	Srithika Kambhammettu	9876543210	sritika@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator: QL File 6* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL Additions

Schemas

Filter objects

bookflow_db

- Tables
 - book1
 - books
 - books1
 - customer_transactions3
 - loans
 - members
 - members1
 - student_marks1
- Views
- Stored Procedures
- Functions
- sys

Administration Schemas

Information

Schema: bookflow_db

Object Info Session

Query Completed

SQL Editor

Execute the selected portion of the script or everything, if there is no selection

(504, 104, 'Personal Loan', 500000, 10.5);

SELECT * FROM Customer;

SELECT * FROM Account;

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

Account 10 x

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	09:41:03	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
19	09:41:03	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
20	09:41:03	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
21	09:41:16	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
22	09:41:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator: QL File 6* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12* SQL Additions

Schemas

Filter objects

bookflow_db

- Tables
 - book1
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 - loans
 - members
 - members1
 - student_marks1
- Views
- Stored Procedures
- Functions
- sys

Administration Schemas

Information

Schema: bookflow_db

Object Info Session

Query Completed

SQL Editor

Execute the selected portion of the script or everything, if there is no selection

SELECT * FROM Customer;

SELECT * FROM Account;

SELECT * FROM Bank_Transaction;

SELECT * FROM Loan;

Result Grid

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-26 09:39:05
2	10002	DEPOSIT	15000.00	2026-08-26 09:39:05
3	10003	WITHDRAW	20000.00	2026-08-26 09:39:05
4	10004	DEPOSIT	5000.00	2026-08-26 09:39:05
5	10005	WITHDRAW	10000.00	2026-08-26 09:39:05

Bank_Transaction 11 x

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	09:41:03	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
20	09:41:03	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
21	09:41:16	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
22	09:41:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
23	09:41:54	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 6* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12*

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

91 • SELECT * FROM Account;

92

93 • SELECT * FROM Bank_Transaction;

94

95 • SELECT * FROM Loan;

96

Result Grid

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50

Loan 12 x

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
20	09:41:03	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
21	09:41:16	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
22	09:41:39	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
23	09:41:54	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
24	09:42:19	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Query Completed

11. PROCEDURE TO DISPLAY ALL CUSTOMERS

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 6* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 12*

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

104

105

106 • -- EXECUTION:

107

108

109 CALL GetAllCustomers();

110

Result Grid

Customer_ID	Customer_Name	Phone	Email	City
101	Srithika Kambhammettu	9876543210	srithika@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

Result 13 x

Read Only

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
24	09:42:19	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
25	10:07:54	EXECUTION: CALL GetAllCustomers()	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
26	10:09:02	CALL GetAllCustomers()	Error Code: 1305. PROCEDURE bankdb.GetAllCustomers does not exist	0.000 sec
27	10:09:42	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	0 row(s) affected	0.015 sec
28	10:09:51	CALL GetAllCustomers()	5 row(s) returned	0.000 sec / 0.000 sec

Query Completed

12. PROCEDURE TO DISPLAY ACCOUNT DETAILS FOR A GIVEN ACCOUNT NUMBER

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema with various tables and views. The main editor window contains the following SQL code:

```
121
122 DELIMITER ;
123
124 -- EXECUTION:
125
126 CALL GetAccountDetails(10001);
127
```

The 'Result Grid' shows the output of the procedure call:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

The 'Output' tab at the bottom shows the execution log:

#	Time	Action	Message	Duration / Fetch
30	10:11:51	BEGIN SELECT * FROM Account WHERE Account_No = p_Account_No	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.000 sec
31	10:12:15	CALL GetAccountDetails(10001)	Error Code: 1305. PROCEDURE bankdb.GetAccountDetails does not exist	0.000 sec
32	10:13:10	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.000 sec
33	10:13:17	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT ...	0 row(s) affected	0.000 sec
34	10:13:30	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec

13. PROCEDURE TO FIND CUSTOMER ACCOUNTS

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema. The main editor window contains the following SQL code:

```
147
148 END //
149
150 DELIMITER ;
151 CALL GetCustomerAccounts(101);
152
153
```

The 'Result Grid' shows the output of the procedure call:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Srithika Kambhammettu	10001	Savings	50000.00	Hyderabad

The 'Output' tab at the bottom shows the execution log:

#	Time	Action	Message	Duration / Fetch
39	08:24:08	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELE...	Error Code: 1304. PROCEDURE GetCustomerAccounts already exists	0.000 sec
40	08:25:05	DROP PROCEDURE IF EXISTS GetCustomerAccounts	0 row(s) affected	0.016 sec
41	08:25:05	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELE...	0 row(s) affected	0.000 sec
42	10:53:43	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELE...	Error Code: 1304. PROCEDURE GetCustomerAccounts already exists	0.000 sec
43	10:57:20	CALL GetCustomerAccounts(101)	1 row(s) returned	0.000 sec / 0.000 sec

14. PROCEDURE TO DEPOSIT MONEY

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains the following code:

```
154 CREATE PROCEDURE DepositMoney(  
155     IN p_Account_No INT,  
156     IN p_Amount DECIMAL(12,2)  
157 )  
158 BEGIN  
159  
160     UPDATE Account  
161     SET Balance = Balance + p_Amount  
162     WHERE Account_No = p_Account_No;  
163  
164 END //  
165  
166 DELIMITER ;  
167 CALL DepositMoney(10001, 5000);  
168
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
43	10:57:20	CALL GetCustomerAccounts(101)	1 row(s) returned	0.000 sec / 0.000 sec
44	10:58:11	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMA...	0 row(s) affected	0.000 sec
45	10:58:31	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.015 sec
46	10:58:46	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMA...	Error Code: 1304. PROCEDURE DepositMoney already exists	0.000 sec
47	11:00:23	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.000 sec

15. PROCEDURE TO WITHDRAW MONEY

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains the following code:

```
170 CREATE PROCEDURE WithdrawMoney(  
171     IN p_Account_No INT,  
172     IN p_Amount DECIMAL(12,2)  
173 )  
174 BEGIN  
175  
176     UPDATE Account  
177     SET Balance = Balance - p_Amount  
178     WHERE Account_No = p_Account_No;  
179  
180 END //  
181  
182 DELIMITER ;  
183 CALL WithdrawMoney(10001, 3000);  
184
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
45	10:58:31	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.015 sec
46	10:58:46	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMA...	Error Code: 1304. PROCEDURE DepositMoney already exists	0.000 sec
47	11:00:23	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.000 sec
48	11:01:37	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIM...	0 row(s) affected	0.000 sec
49	11:02:06	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.000 sec

TRIGGERS

16. TRIGGER – PREVENT NEGATIVE BALANCE

The screenshot shows the MySQL Workbench interface with a SQL script in the editor. The script defines a trigger named 'CheckBalance' that fires BEFORE UPDATE ON Account. The trigger logic checks if the new balance is less than zero. If so, it signals an SQLSTATE '45000', sets a message text 'Transaction failed: Insufficient balance', and ends the trigger. The script also includes an UPDATE statement for Account_No = 10001.

```
191 IF NEW.Balance < 0 THEN
192
193     SIGNAL SQLSTATE '45000'
194     SET MESSAGE_TEXT =
195         'Transaction failed: Insufficient balance';
196
197 END IF;
198
199 END //
200
201 DELIMITER ;
202
203 UPDATE Account
204 SET Balance = Balance - 60000
205 WHERE Account_No = 10001;
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
47	11:00:23	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.000 sec
48	11:01:37	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIM...	0 row(s) affected	0.000 sec
49	11:02:06	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.000 sec
50	11:04:18	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEG...	0 row(s) affected	0.016 sec
51	11:04:36	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance	0.015 sec

Query interrupted

17. TRIGGER – PREVENT NEGATIVE DEPOSIT

The screenshot shows the MySQL Workbench interface with a SQL script in the editor. The script defines a trigger named 'CheckTransactionAmount' that fires BEFORE INSERT ON Bank_Transaction. The trigger logic checks if the transaction amount is less than or equal to zero. If so, it signals an SQLSTATE '45000', sets a message text 'Transaction amount must be greater than zero', and ends the trigger. The script also includes an INSERT statement for Bank_Transaction.

```
215 SET MESSAGE_TEXT =
216     'Transaction amount must be greater than zero';
217
218 END IF;
219
220 END //
221
222 DELIMITER ;
223
224 INSERT INTO Bank_Transaction
225 (Account_No, Transaction_Type, Amount)
226 VALUES
227 (10001, 'DEPOSIT', -5000);
228
229
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
49	11:02:06	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.000 sec
50	11:04:18	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEG...	0 row(s) affected	0.016 sec
51	11:04:36	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance	0.015 sec
52	11:05:35	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FO...	0 row(s) affected	0.000 sec
53	11:05:56	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec

Query interrupted

18. CREATE TRANSACTION AUDIT TABLE

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains the following code:

```
DELIMITER ;
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', -5000);
CREATE TABLE Transaction_Audit (
  Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
  Transaction_ID INT,
  Account_No INT,
  Transaction_Type VARCHAR(20),
  Amount DECIMAL(12,2),
  Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
50	11:04:18	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEG...	0 row(s) affected	0.016 sec
51	11:04:36	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance	0.015 sec
52	11:05:35	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FO...	0 row(s) affected	0.000 sec
53	11:05:56	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec
54	11:07:01	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected	0.031 sec

19. TRIGGER – TRANSACTION AUDIT

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains the following code:

```
DELIMITER ;
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 2500);
SELECT * FROM Transaction_Audit;
```

The Output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
53	11:05:56	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec
54	11:07:01	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected	0.031 sec
55	11:09:22	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH R...	0 row(s) affected	0.016 sec
56	11:10:29	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	1 row(s) affected	0.015 sec
57	11:10:55	SELECT * FROM Transaction_Audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The Result Grid shows the data from the Transaction_Audit table:

Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
1	6	10001	DEPOSIT	2500.00	2026-08-27 11:10:29

20. TRIGGER – AUTOMATICALLY UPDATE ACCOUNT BALANCE

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains a query to select account information for account number 10001. The output grid shows the result of the query.

SQL Editor:

```
293 (Account_No, Transaction_Type, Amount)
294 VA Execute the selected portion of the script or everything, if there is no selection
295 (10001, 'DEPOSIT', 5000);
296 SELECT *
297 FROM Account
298 WHERE Account_No = 10001;
299
```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	62000.00	Hyderabad

Output:

#	Time	Action	Message	Duration / Fetch
57	11:10:55	SELECT * FROM Transaction_Audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
58	11:12:23	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; database exists	0.000 sec
59	11:12:31	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction...	0 row(s) affected	0.000 sec
60	11:13:03	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	1 row(s) affected	0.015 sec
61	11:13:22	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Query Completed

21. TRIGGER – PREVENT WITHDRAWAL ABOVE BALANCE

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains a query to insert a withdrawal transaction for account number 10001. The output grid shows the result of the query, with errors indicating that the withdrawal failed due to insufficient balance.

SQL Editor:

```
318 Execute the selected portion of the script or everything, if there is no selection
319
320 END IF;
321
322 END //
323
324 DELIMITER ;
325 INSERT INTO Bank_Transaction
326 (Account_No, Transaction_Type, Amount)
327 VALUES
328 (10001, 'WITHDRAW', 1000000);
329
330
331
332
```

Output:

#	Time	Action	Message	Duration / Fetch
63	11:15:41	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec
64	11:15:59	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec
65	11:17:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	1 row(s) affected	0.015 sec
66	11:18:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
67	11:19:29	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec

Query interrupted

ADVANCED STORED PROCEDURES

22. PROCEDURE – ACCOUNT TRANSFER

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains the following code:

```
365 DELIMITER ;
366 CALL TransferMoney(10001, 10002, 5000);
367 SELECT *
368 FROM Account
369 WHERE Account_No IN (10001,10002);
370
371
```

The 'Result Grid' displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	56900.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
66	11:18:27	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
67	11:19:29	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1000...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec
68	11:21:26	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount I...	0 row(s) affected	0.000 sec
69	11:21:40	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected	0.000 sec
70	11:21:53	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

23. PROCEDURE – CUSTOMER LOAN DETAILS

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains the following code:

```
387
388 END //
389
390 DELIMITER ;
391 CALL GetCustomerLoans(101);
392
393
```

The 'Result Grid' displays the following data:

Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
Srithika Kamthammettu	501	Home Loan	5000000.00	7.50

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
68	11:21:26	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount I...	0 row(s) affected	0.000 sec
69	11:21:40	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected	0.000 sec
70	11:21:53	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
71	11:22:28	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT ...	0 row(s) affected	0.015 sec
72	11:22:42	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec

24. PROCEDURE – HIGH BALANCE ACCOUNTS

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'bookflow_db' selected. The main editor window shows a SQL script with the following content:

```
403  
404 END //  
405  
406 DELIMITER ;  
407 CALL HighBalanceAccounts(50000);  
408  
409
```

The 'Result Grid' is displayed below the script, showing the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada
10001	101	Savings	56900.00	Hyderabad

The 'Output' pane at the bottom shows the execution log:

#	Time	Action	Message	Duration / Fetch
70	11:21:53	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
71	11:22:28	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT ...	0 row(s) affected	0.015 sec
72	11:22:42	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec
73	11:24:38	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2)) BE...	0 row(s) affected	0.000 sec
74	11:26:27	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec

25. PROCEDURE WITH IN, OUT PARAMETERS

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'bookflow_db' selected. The main editor window shows a SQL script with the following content:

```
420  
421  
422  
423 DELIMITER ;  
424  
425 CALL GetBalance(10001, @CurrentBalance);  
426
```

The 'Result Grid' is displayed below the script, showing the following data:

@CurrentBalance
56900.00

The 'Output' pane at the bottom shows the execution log:

#	Time	Action	Message	Duration / Fetch
73	11:24:38	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2)) BE...	0 row(s) affected	0.000 sec
74	11:26:27	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec
75	11:26:59	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMA...	0 row(s) affected	0.000 sec
76	11:27:20	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected	0.000 sec
77	11:27:20	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

TRIGGER MANAGEMENT COMMANDS

MySQL Workbench interface showing the execution of trigger management commands. The SQL editor contains the following commands:

```
426
427 • SELECT @CurrentBalance;
428 • SHOW TRIGGERS;
429
430
431 • SHOW CREATE TRIGGER CheckBalance;
432
```

The Output pane displays the results of these commands:

#	Time	Action	Message	Duration / Fetch
74	11:26:27	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec
75	11:26:59	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMA...	0 row(s) affected	0.000 sec
76	11:27:20	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected	0.000 sec
77	11:27:20	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
78	12:25:03	SHOW TRIGGERS	5 row(s) returned	0.015 sec / 0.000 sec

MySQL Workbench interface showing the execution of the `SHOW CREATE TRIGGER` command. The SQL editor contains the following commands:

```
427 • SELECT @CurrentBalance;
428 • SHOW TRIGGERS;
429
430
431 • SHOW CREATE TRIGGER CheckBalance;
432
```

The Output pane displays the results of these commands:

#	Time	Action	Message	Duration / Fetch
75	11:26:59	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMA...	0 row(s) affected	0.000 sec
76	11:27:20	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected	0.000 sec
77	11:27:20	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
78	12:25:03	SHOW TRIGGERS	5 row(s) returned	0.015 sec / 0.000 sec
79	12:25:20	SHOW CREATE TRIGGER CheckBalance	1 row(s) returned	0.015 sec / 0.000 sec

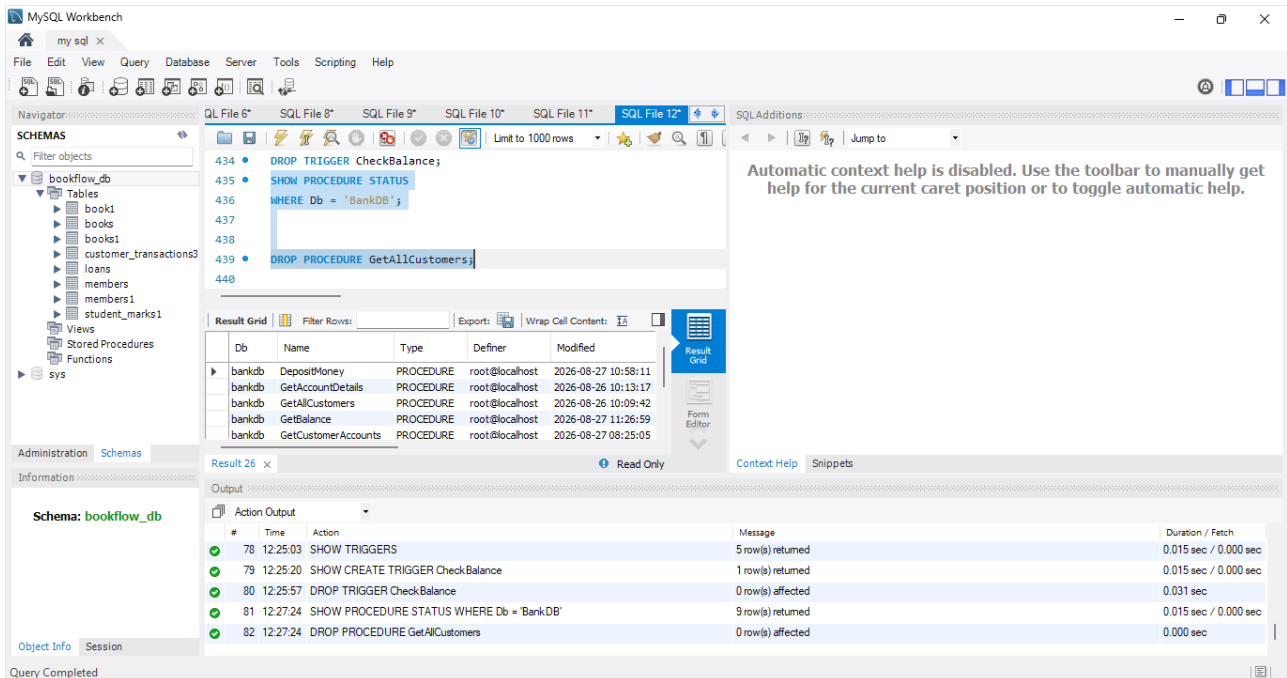
MySQL Workbench interface showing the execution of the `DROP TRIGGER` command. The SQL editor contains the following commands:

```
426
427 • SELECT @CurrentBalance;
428 • SHOW TRIGGERS;
429
430
431 • SHOW CREATE TRIGGER CheckBalance;
432
433
434 • DROP TRIGGER CheckBalance;
435
```

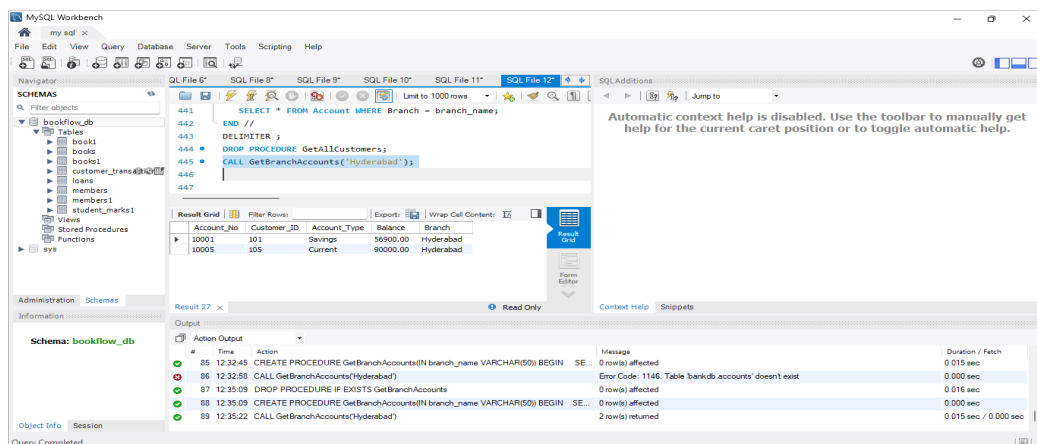
The Output pane displays the results of these commands:

#	Time	Action	Message	Duration / Fetch
76	11:27:20	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected	0.000 sec
77	11:27:20	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
78	12:25:03	SHOW TRIGGERS	5 row(s) returned	0.015 sec / 0.000 sec
79	12:25:20	SHOW CREATE TRIGGER CheckBalance	1 row(s) returned	0.015 sec / 0.000 sec
80	12:25:57	DROP TRIGGER CheckBalance	0 row(s) affected	0.031 sec

PROCEDURE MANAGEMENT COMMANDS

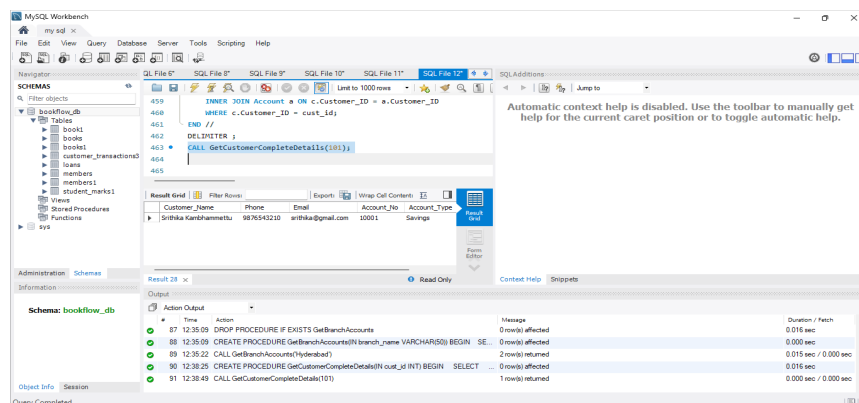


PRACTICE QUESTIONS



QUESTION 2:

Create a procedure that accepts Customer_ID and displays:



QUESTION 3:

Create a procedure to calculate the total balance of all accounts belonging to a particular customer.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
473 WHERE Customer_ID = cust_id
474 GROUP BY Customer_ID;
475
476 END //
477 DELIMITER ;
478 CALL GetTotalCustomerBalance(101);
479
```

The Result Grid shows the output of the query:

Customer_ID	Total_Combined_Balance
101	56900.00

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
89	12:35:22	CALL GetBranchAccounts(Hyderabad)	2 row(s) returned	0.015 sec / 0.000 sec
90	12:38:25	CREATE PROCEDURE GetCustomerCompleteDetails(IN cust_id INT) BEGIN SELECT ...	0 row(s) affected	0.016 sec
91	12:38:49	CALL GetCustomerCompleteDetails(101)	1 row(s) returned	0.000 sec / 0.000 sec
92	12:40:50	CREATE PROCEDURE GetTotalCustomerBalance(IN cust_id INT) BEGIN SELECT ...	0 row(s) affected	0.015 sec
93	12:41:19	CALL GetTotalCustomerBalance(101)	1 row(s) returned	0.000 sec / 0.000 sec

QUESTION 4:

Create a procedure to display accounts whose balance is greater than a given amount.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
483 FROM Account
484
485 END //
486 DELIMITER ;
487
488 CALL HighBalanceAccounts(100000);
489
```

The Result Grid shows the output of the query:

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
94	13:44:10	CREATE PROCEDURE HighBalanceAccounts(IN min_balance DECIMAL(15,2)) BEGIN ...	Error Code: 1304. PROCEDURE HighBalanceAccounts already exists	0.000 sec
95	13:45:07	CALL HighBalanceAccounts(100000)	1 row(s) returned	0.000 sec / 0.000 sec
96	13:46:31	DROP PROCEDURE IF EXISTS HighBalanceAccounts	0 row(s) affected	0.015 sec
97	13:46:31	CREATE PROCEDURE HighBalanceAccounts(IN min_balance DECIMAL(15,2)) BEGIN ...	0 row(s) affected	0.016 sec
98	13:46:38	CALL HighBalanceAccounts(100000)	1 row(s) returned	0.000 sec / 0.000 sec

QUESTION 5:

Create a trigger that prevents an account balance from becoming negative.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'bookflow_db' selected. The main editor window shows the following SQL code:

```
494 CREATE TRIGGER CheckNegativeBalance
495 BEFORE UPDATE ON Account
496 FOR EACH ROW
497 BEGIN
498     IF NEW.Balance < 0 THEN
499         SIGNAL SQLSTATE '45000'
500         SET MESSAGE_TEXT = 'Transaction cancelled: Balance cannot be negative';
501     END IF;
502 END //
503 DELIMITER ;
504
505 -- Step 1: Try to force a negative balance (This will fail)
506 UPDATE Account SET Balance = -500 WHERE Account_No = 10001;
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." The bottom output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
99	13:48:30	CREATE DATABASE BankDB	Error Code: 1007. Can't create database 'bankdb'; database exists	0.000 sec
100	13:48:59	UPDATE Account SET Balance = -500 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
101	13:49:20	DROP TRIGGER IF EXISTS CheckNegativeBalance	0 row(s) affected. 1 warning(s): 1360 Trigger does not exist	0.000 sec
102	13:49:20	CREATE TRIGGER CheckNegativeBalance BEFORE UPDATE ON Account FOR EACH ROW	0 row(s) affected	0.016 sec
103	13:49:52	UPDATE Account SET Balance = -500 WHERE Account_No = 10001	Error Code: 1644. Transaction cancelled: Balance cannot be negative!	0.000 sec

Query interrupted

QUESTION 6:

Create a trigger that prevents a transaction amount from being zero or negative.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'bookflow_db' selected. The main editor window shows the following SQL code:

```
507 UPDATE Account SET Balance = -500 WHERE Account_No = 10001;
508 DELIMITER //
509
510 CREATE TRIGGER bookflow_db.before_transaction_insert
511 BEFORE INSERT ON bookflow_db.customer_transactions3
512 FOR EACH ROW
513 BEGIN
514     IF NEW.amount <= 0 THEN
515         SIGNAL SQLSTATE '45000'
516         SET MESSAGE_TEXT = 'Transaction amount must be greater than zero.';
517     END IF;
518 END//
519
520 DELIMITER ;
521
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." The bottom output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
117	10:16:22	CREATE TRIGGER before_transaction_update BEFORE UPDATE ON customer_transac...	Error Code: 1146. Table 'bankdb.customer_transactions3' doesn't exist	0.000 sec
118	10:17:02	CREATE TRIGGER before_transaction_update BEFORE UPDATE ON customer_transac...	Error Code: 1146. Table 'bankdb.customer_transactions3' doesn't exist	0.000 sec
119	10:17:39	CREATE TRIGGER before_transaction_update BEFORE UPDATE ON bookflow_db.custo...	Error Code: 1435. Trigger in wrong schema	0.015 sec
120	10:18:21	CREATE TRIGGER bookflow_db.before_transaction_update BEFORE UPDATE ON bookf...	0 row(s) affected	0.031 sec
121	10:19:14	CREATE TRIGGER bookflow_db.before_transaction_insert BEFORE INSERT ON bookf...	0 row(s) affected	0.016 sec

Query Completed

QUESTION 7:

Create a trigger that automatically records every transaction in Transaction_Audit.

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains the following code:

```
543      action_performed
544      Execute the selected portion of the script or everything, if there is no selection
545      VALUES (
546          NEW.txn_id,      -- Maps to your transaction ID column
547          NEW.Account_No,  -- Maps to your Account Number column
548          NEW.Transaction_Type, -- Maps to your Type column
549          NEW.amount,      -- Maps to your Amount column
550      );
551      'INSERT'
552      END//
553
554      DELIMITER ;
555
556
557
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
119	10:17:39	CREATE TRIGGER before_transaction_update BEFORE UPDATE ON bookflow_db.custo...	Error Code: 1435. Trigger in wrong schema	0.015 sec
120	10:18:21	CREATE TRIGGER bookflow_db.before_transaction_update BEFORE UPDATE ON bookfl...	0 row(s) affected	0.031 sec
121	10:19:14	CREATE TRIGGER bookflow_db.before_transaction_insert BEFORE INSERT ON bookflow...	0 row(s) affected	0.016 sec
122	10:22:15	CREATE TABLE IF NOT EXISTS bookflow_db.Transaction_Audit (audit_id INT AUTO_I...	0 row(s) affected	0.015 sec
123	10:22:44	CREATE TRIGGER bookflow_db.after_transaction_insert AFTER INSERT ON bookflow_db...	0 row(s) affected	0.000 sec

QUESTION 8:

Create a trigger that automatically increases the account balance whenever a DEPOSIT transaction is inserted.

The screenshot shows the MySQL Workbench interface with the 'bookflow_db' schema selected. The SQL editor contains the following code:

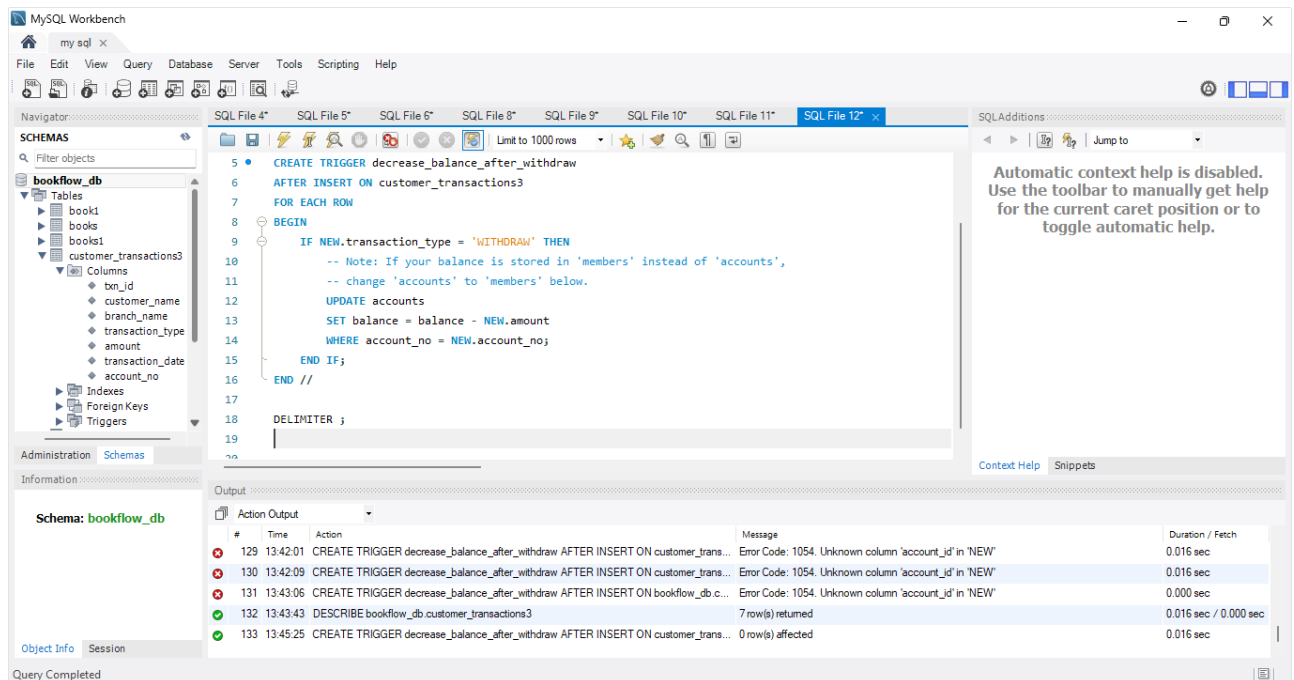
```
562      Note: Adjust the string "DEPOSIT" if your database uses lowercase or short codes
563      IF UPPER(NEW.Transaction_Type) = 'DEPOSIT' THEN
564
565          UPDATE bookflow_db.Account
566          SET Balance = Balance + NEW.amount
567          WHERE Account_No = NEW.Account_No;
568
569      END IF;
570
571      END//
572
573      DELIMITER ;
574
575
576
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
120	10:18:21	CREATE TRIGGER bookflow_db.before_transaction_update BEFORE UPDATE ON bookfl...	0 row(s) affected	0.031 sec
121	10:19:14	CREATE TRIGGER bookflow_db.before_transaction_insert BEFORE INSERT ON bookflow...	0 row(s) affected	0.016 sec
122	10:22:15	CREATE TABLE IF NOT EXISTS bookflow_db.Transaction_Audit (audit_id INT AUTO_I...	0 row(s) affected	0.015 sec
123	10:22:44	CREATE TRIGGER bookflow_db.after_transaction_insert AFTER INSERT ON bookflow_db...	0 row(s) affected	0.000 sec
124	10:23:44	CREATE TRIGGER bookflow_db.after_deposit_insert AFTER INSERT ON bookflow_db.cu...	0 row(s) affected	0.000 sec

QUESTION 9:

Create a trigger that automatically decreases the account balance whenever a **WITHDRAW** transaction is inserted.



QUESTION 10:

Create a trigger that prevents withdrawal when the account does not have sufficient balance.

